

JKBOSE 2026

CLASS 11TH

STATISTICS

MATHS

LECTURE-1

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TOPICS TO BE COVERED

1

Introduction to Statistics

2

Measures of Dispersion

3

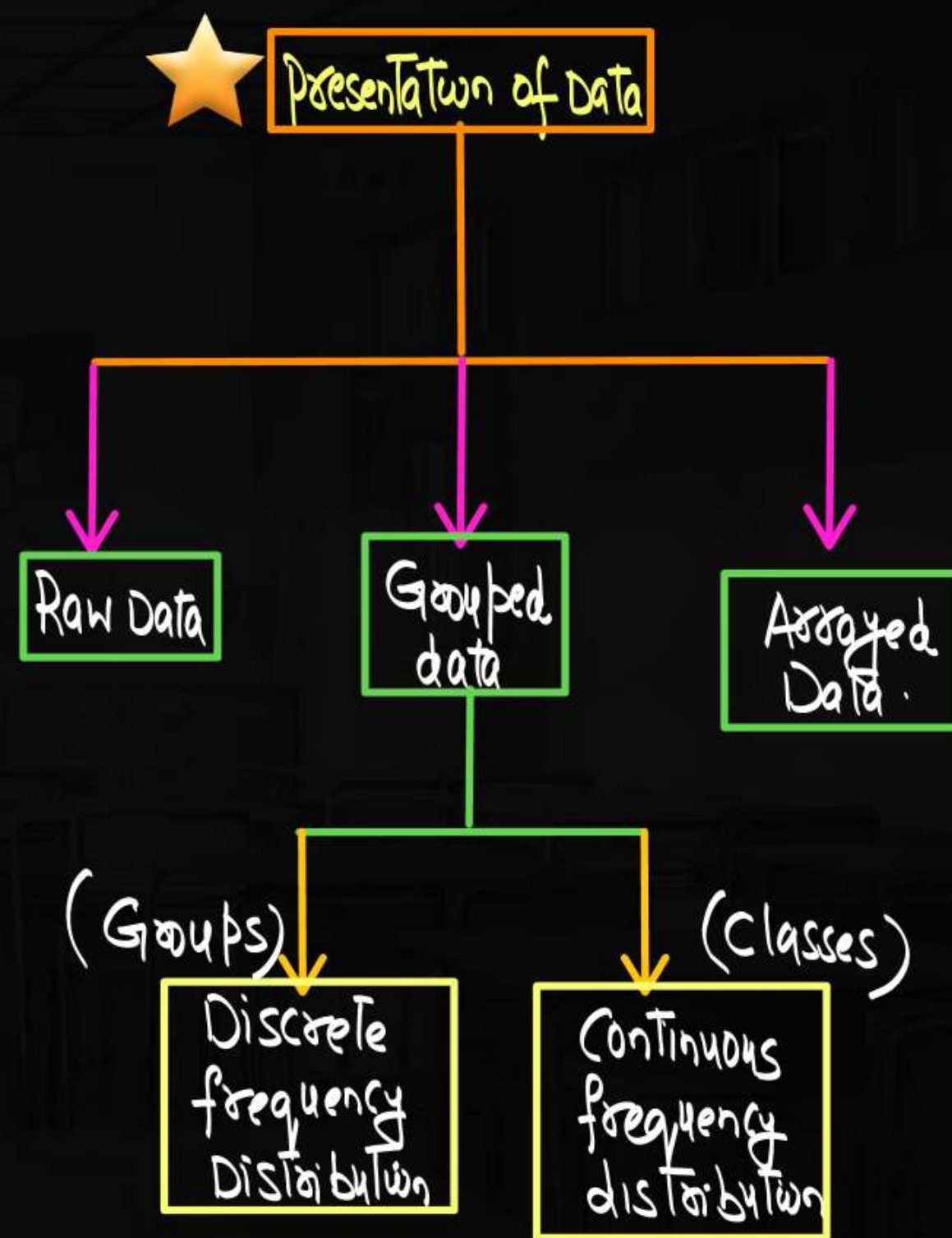
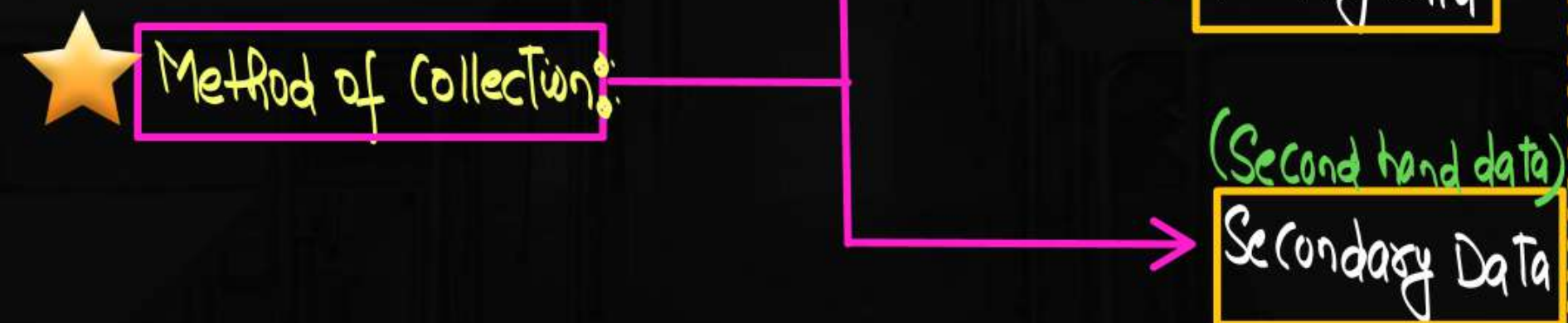
Related Problems

INTRODUCTION

★ **Statistics**: The Science of collecting, organising analysing and Interpreting of Data is known as Statistics.

★ **Data**: Collection of numerical figures representing a particular information is known as data.

* Data Can be classified on the basis of the method of collection & presentation of Data. (first hand data)



Primary Data

- Data collected by an individual or by a group of investigators directly, is called a Primary data.
- Its also called the first hand data.
Example: Data collected by a five students about the height of 50 plants in their school.

Secondary data

- Data collected from a source which already had the information stored, is called a Secondary data
- Its also called the Second hand data.
Example: Data obtained from internet, Books, Magazines etc.

Raw Data

- Initial collection of observations, which is not following any order is called raw data.

Example: The marks obtained by 10 students in a monthly test (10, 99, 50, 64, 66, 90, 97, 80, 72, 6)

Grouped data

- In order to present data in a more meaningful way, we condense the data into classes or groups. This form of data is called grouped data.

Example:

Height of students	5'0"	5'2"	5'4"	5'6"
Number of students	6	2	5	4

(Discrete frequency Distribution)

Arrayed data

- Data arranged in increasing order or in decreasing order is said to be an arrayed data.

QUESTION

Primary data can be collected from

- 1 Books ✗
- 2 Historical facts ✗
- 3 Interviews ✓✓
- 4 Magazines ✗

QUESTION

The given data : 5, 8, 7, 6, 4, 2, 9, 7, 3, 1, 2, 4 is

1 Raw data ✓

2 Arrayed data ✗

3 Primary data ✗

4 None of these ✗

QUESTION

The weight of 10 students is : 14, 16, 18, 19, 20, 21, 22, 28, 29, 30. The given data corresponds to

1 Raw data ✗

2 Arrayed data ✓✓

3 Primary data ✗

4 Secondary data ✗

QUESTION

The data corresponds to

(Discrete frequency Distribution)

Marks	50	60	89	90	95
Number of Students	4	3	7	6	2

1 Raw data ✗

2 Arrayed data ✗

3 Primary data ✗

4 Grouped data ✓✓

Frequency

- The **number of observations** in a **particular group or class** is called frequency of that group.

Weights	Tally marks	Frequency
15-25	I	1
25-35	IIII IIII	9
35-45	IIII II	7
45-55	IIII	4
55-65	IIII IIII IIII I	16

(Continuous frequency Distribution)

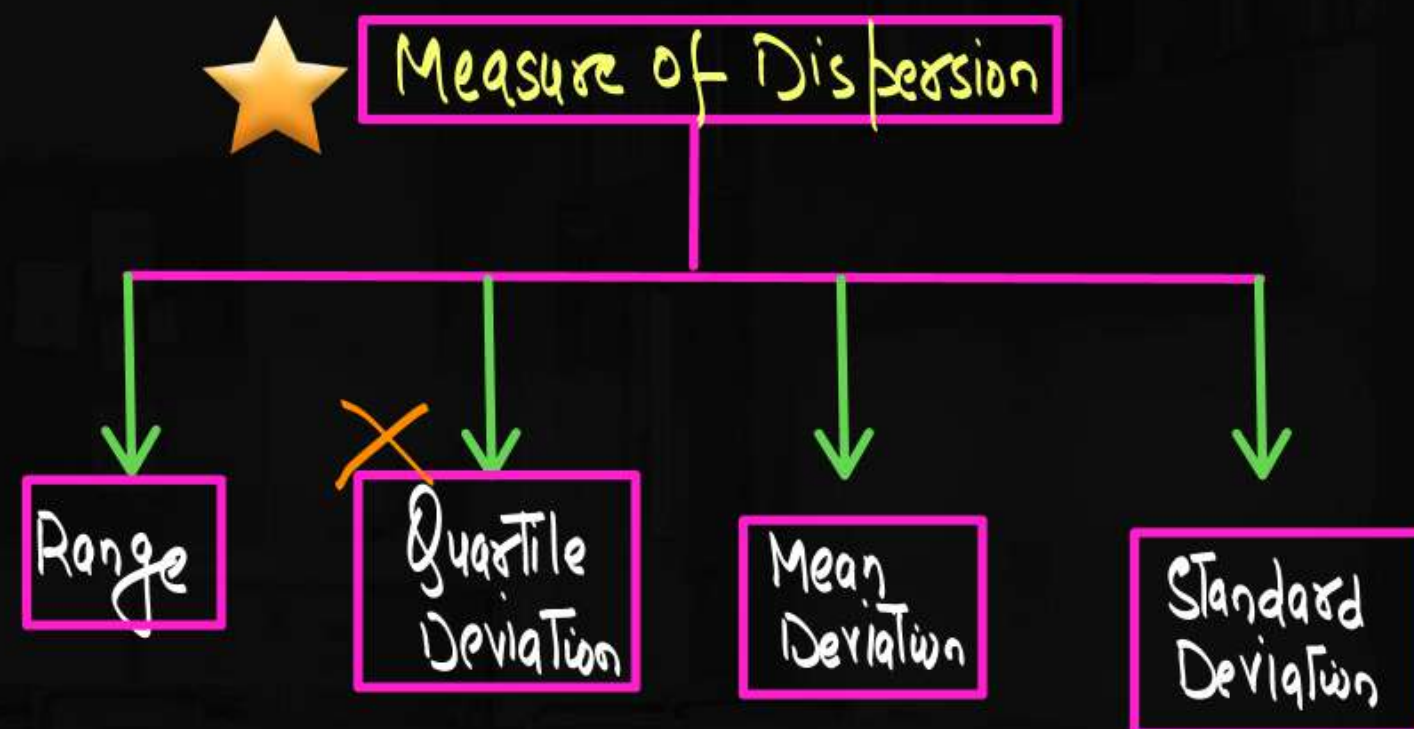
Measures of Dispersion

- * It is a statistical tool used to Measure the spread or scatteredness of the data set.
- * It tells us how much the data values differ from each other and from their average (Mean/Median).

* For the Measure of the dispersion, we have the following Tools:

- 1) Range $\rightarrow$ (Max. value - Min. value)
- 2) Quartile Deviation ^X
- (3) Mean Deviation $\rightarrow$ (Deviation from Mean or Median)

(4) Standard Deviation.



Mean deviation for ungrouped data

Suppose that $x_1, x_2, x_3, \dots, x_n$ be the given observation

& let a represents their Average (Mean/Median).

Then, we proceed as follows:

(1) Calculate the Deviations from a :

$$x_1 - a, x_2 - a, x_3 - a, \dots, x_n - a.$$

(2) Take the absolute value of the deviations:

$$|x_1 - a|, |x_2 - a|, \dots, |x_n - a|.$$

(3) The mean of the absolute value of the deviations represents the Mean deviation from a .

$$\text{So, } M.D.(a) = \frac{|x_1 - a| + |x_2 - a| + \dots + |x_n - a|}{n}$$

$$\Rightarrow M.D.(a) = \frac{\sum_{i=1}^n |x_i - a|}{n}$$

$$\text{Here } M.D(\bar{x}) = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n} \quad (\text{Mean Deviation from Mean})$$

$$\& M.D.(M) = \frac{\sum_{i=1}^n |x_i - M|}{n} \quad (\text{Mean Deviation from Median})$$

EXAMPLE

Find the **mean deviation** about the **mean** for the following data:

6, 7, 10, 12, 13, 4, 8, 12

(Sol) Here $\bar{x} = \frac{6+7+10+12+13+4+8+12}{8}$

$\Rightarrow \bar{x} = \frac{72}{8} = 9 \Rightarrow \bar{x} = 9$

Now, the absolute value of Deviations is given as follows:

$|6-9|, |7-9|, |10-9|, |12-9|, |13-9|, |4-9|, |8-9|, |12-9|$
 i.e. 3, 2, 1, 3, 4, 5, 1, 3.

So $M.D(\bar{x}) = \frac{3+2+1+3+4+5+1+3}{8}$

$\Rightarrow M.D(\bar{x}) = \frac{22}{8}$

$\Rightarrow M.D(\bar{x}) = 2.75$

EXAMPLE

Find the **mean deviation** about the ^{Median}~~mean~~ for the following data:

3, 9, 5, 3, 12, 10, 18, 4, 7, 19, 21.

(Sol) Arrange the given in the ascending order:

3, 3, 4, 5, 7, 9, 10, 12, 18, 19, 21

Here $n = 11$, which is odd. So Median is given by:

$$M = T_{\frac{n+1}{2}} = T_6 =$$

$$\Rightarrow M = 9$$

$$M = \frac{1}{2} \left(T_{\frac{n}{2}} + T_{\frac{n}{2}+1} \right)$$

$$\text{Therefore } M.D(M) = \frac{6+6+5+4+2+0+1+3+9+10+12}{11}$$

$$\Rightarrow M.D(M) = \frac{58}{11}$$

$$\Rightarrow M.D(M) = 5.27$$

QUESTION (NCERT EXAMPLER)

The mean deviation of data 2, 9, 9, 3, 6, 9, 4 from the mean is

1 ✗ 2.23

2 ✓ 2.57

3 ✗ 3.23

4 ✗ 3.57

$$\text{(Sol)} \text{ Here } \bar{x} = \frac{2+9+9+3+6+9+4}{7} = \frac{42}{7}$$

$$\Rightarrow \bar{x} = 6$$

$$\text{So M.D}(\bar{x}) = \frac{4+3+3+3+0+3+2}{7}$$

$$\Rightarrow \text{M.D}(\bar{x}) = \frac{18}{7} = 2.57$$

QUESTION (NCERT EXAMPLER)

Mean deviation for n observations $x_1, x_2, \dots, x_n$ from their mean $\bar{x}$ is given by

1 $\sum_{i=1}^n (x_i - \bar{x})$ ✗

2 $\frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$ ✓

3 $\sum_{i=1}^n (x_i - \bar{x})^2$ ✗

4 $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ ✗

QUESTION (NCERT EXAMPLER)

Following are the marks obtained by 9 students in a mathematics test:

50, 69, 20, 33, 53, 39, 40, 65, 59. The mean deviation from the median is:

- | | | |
|----------|---------|--|
| 1 | 9 ✗ | <p>(Sol) Ascending order:
 $20, 33, 39, 40, 50, 53, 59, 65, 69.$
 So $M = T_{\frac{n+1}{2}} = T_{\frac{9+1}{2}} = T_5 = 50.$</p> |
| 2 | 10.5 ✗ | |
| 3 | 12.67 ✓ | <p>$\Rightarrow M.D.(M) = \frac{30 + 17 + 11 + 10 + 0 + 3 + 9 + 15 + 19}{9}$</p> |
| 4 | 14.76 ✗ | |

$$M.D.(M) = 12.67$$

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ZAHEEN MANZOODR.



Waleeha Wani



Thank You